Лабораторная работа 6

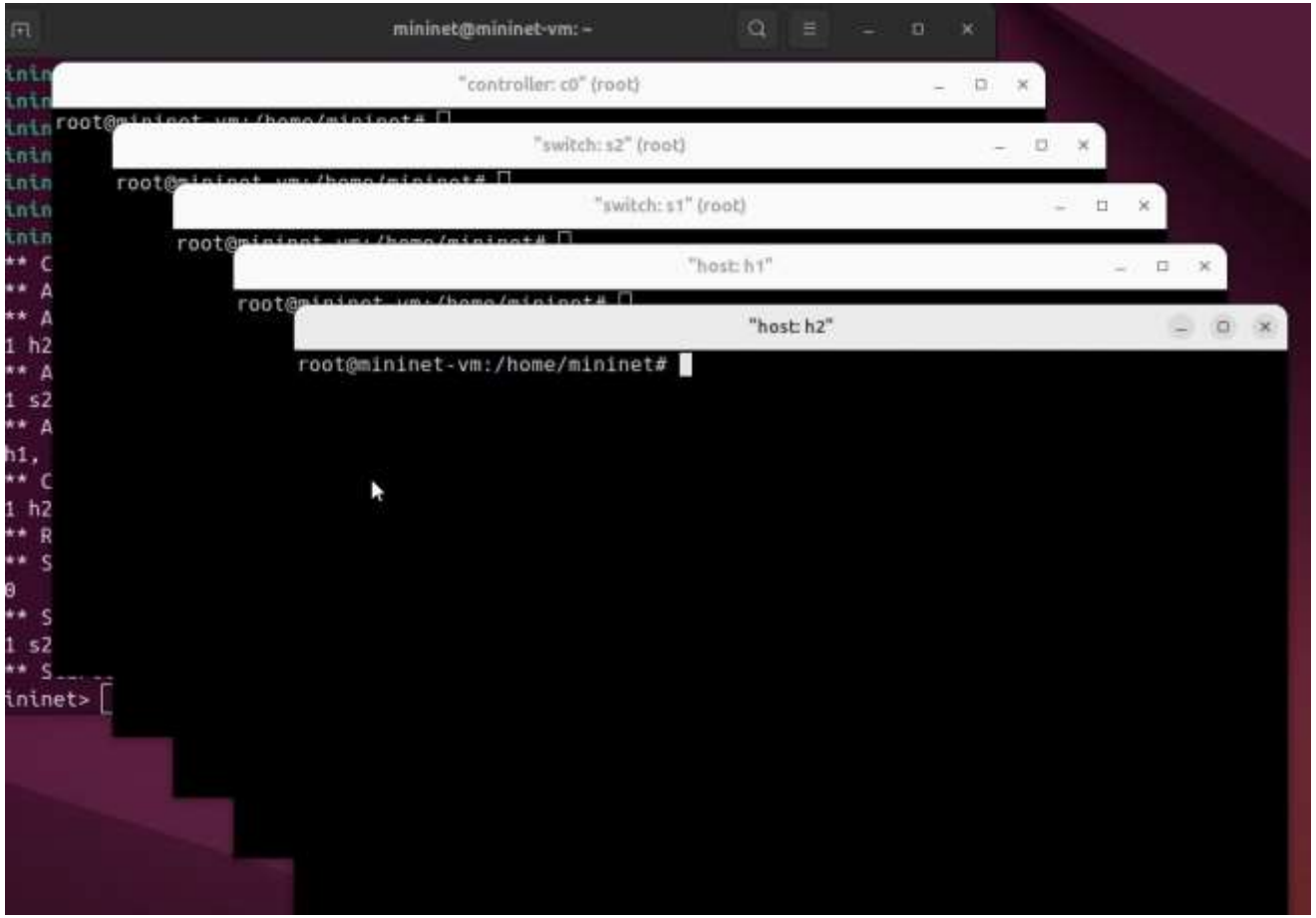
Настройка пропускной способности глобальной сети с помощью Token Bucket Filter

Цвелев С.А. НПИбд-01-22

Цель работы

Основной целью работы является знакомство с принципами работы дисциплины очереди Token Bucket Filter, которая формирует входящий/исходящий трафик для ограничения пропускной способности, а также получение навыков моделирования и исследования поведения трафика посредством проведения интерактивного и воспроизводимого экспериментов в Mininet.

Создание базовой топологии



Интерактивные эксперименты по ограничению пропускной способности сети

```
"host: h1"
root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
Connecting to host 10.0.0.2, port 5201
[  7] local 10.0.0.1 port 40954 connected to 10.0.0.2 port 5201
[ ID] Interval           Transfer     Bitrate         Retr  Cwnd
[  7]   0.00-1.00   sec  3.66 GBytes  31.4 Gbits/sec  995    771 KBytes
[  7]   1.00-2.00   sec  3.60 GBytes  30.9 Gbits/sec  1747    665 KBytes
[  7]   2.00-3.00   sec  3.48 GBytes  29.9 Gbits/sec  680   1.25 MBytes
[  7]   3.00-4.00   sec  2.95 GBytes  25.3 Gbits/sec  1090    932 KBytes
[  7]   4.00-5.00   sec  3.72 GBytes  31.9 Gbits/sec  2878   1.17 MBytes
[  7]   5.00-6.00   sec  3.86 GBytes  33.2 Gbits/sec  1073    830 KBytes
[  7]   6.00-7.00   sec  3.79 GBytes  32.6 Gbits/sec  1046   1.70 MBytes
[  7]   7.00-8.00   sec  3.72 GBytes  32.0 Gbits/sec  351    809 KBytes
[  7]   8.00-9.00   sec  4.08 GBytes  35.0 Gbits/sec  2350    990 KBytes
[  7]   9.00-10.00  sec  3.85 GBytes  33.1 Gbits/sec  935   1.94 MBytes
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[ ID] Interval           Transfer     Bitrate         Retr
[  7]   0.00-10.00  sec  36.7 GBytes  31.5 Gbits/sec  13145             sender
[  7]   0.00-10.00  sec  36.7 GBytes  31.5 Gbits/sec                  receiver

iperf Done.
root@mininet-vm:/home/mininet# sudo tc qdisc add dev h1-eth0 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet#
```
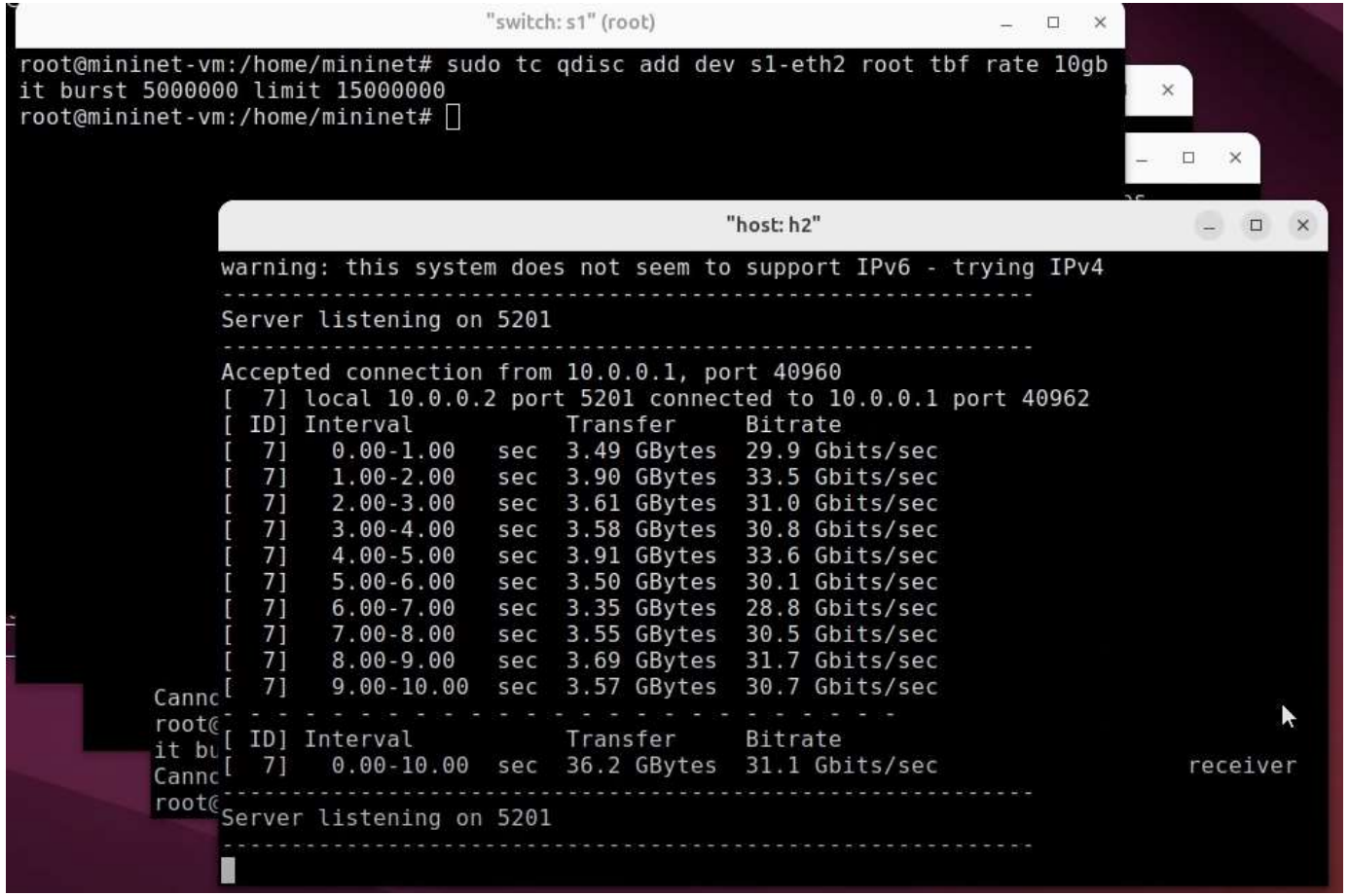

Интерактивные эксперименты по ограничению пропускной способности сети



Интерактивные эксперименты по ограничению пропускной способности сети

```
"switch: s1" (root)
root@mininet-vm:/home/mininet# sudo tc qdisc add dev s1-eth2 root tbf rate 10gb
it burst 5000000 limit 15000000
root@mininet-vm:/home/mininet# suto tc qdisc del dev s1-eth2 root

Command 's
          root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
  command Connecting to host 10.0.0.2, port 5201
  command [  7] local 10.0.0.1 port 40966 connected to 10.0.0.2 port 5201
  command [ ID] Interval           Transfer     Bitrate         Retr  Cwnd
  command [  7]   0.00-1.00   sec  1.11 GBytes  9.54 Gbits/sec  534    860 KBytes
          [  7]   1.00-2.00   sec  1.11 GBytes  9.54 Gbits/sec   61    889 KBytes
Try: apt i[  7]   2.00-3.00   sec  1.11 GBytes  9.54 Gbits/sec  133    755 KBytes
          [  7]   3.00-4.00   sec  1.11 GBytes  9.55 Gbits/sec  392    423 KBytes
root@minir[  7]   4.00-5.00   sec  1.11 GBytes  9.54 Gbits/sec  368    851 KBytes
root@minir^C[  7]   5.00-5.30   sec   317 MBytes  8.90 Gbits/sec    3    935 KBytes
em delay 1- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -
root@minir[ ID] Interval           Transfer     Bitrate         Retr
: tbf rate[  7]   0.00-5.30   sec  5.87 GBytes  9.51 Gbits/sec  1491             sende
root@minir[  7]   0.00-5.30   sec  0.00 Bytes  0.00 bits/sec                  receiver
          iperf3: interrupt - the client has terminated
          root@mininet-vm:/home/mininet# iperf3 -c 10.0.0.2
          Connecting to host 10.0.0.2, port 5201
          [  7] local 10.0.0.1 port 40970 connected to 10.0.0.2 port 5201
          [ ID] Interval           Transfer     Bitrate         Retr  Cwnd
          [  7]   0.00-1.00   sec  42.5 MBytes   356 Mbits/sec   45   1.33 MBytes
          [  7]   1.00-2.01   sec  53.8 MBytes   446 Mbits/sec    0   1.36 MBytes
          [  7]   2.01-3.00   sec  43.8 MBytes   371 Mbits/sec    0   1.38 MBytes
          [  7]   3.00-4.00   sec  18.8 MBytes   157 Mbits/sec    0   1.39 MBytes
```